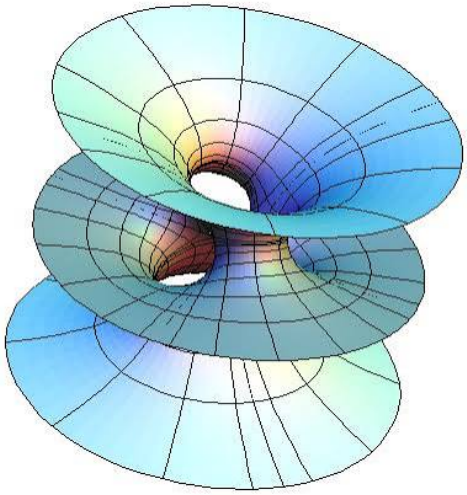


Manifolds

An Intro to Differential Geometry

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1 Manifolds

1.1 Defining a Manifold

Definition 1.1 (Diffeomorphism). Let $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^n$ be open subsets. Then a smooth (infinitely differentiable) map $F : U \to V$ is said to be a *diffeomorphism* if $F^{-1} : V \to U$ is also smooth.

Definition 1.2 (Jacobian). The *Jacobian matrix* $DF(\mathbf{x})$ of a smooth map $F : U \to V$ between open subsets $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^n$ for $\mathbf{x} \in U$ is defined as the $n \times m$ matrix with following entries:

$$(DF(\mathbf{x}))^i_j = \frac{\partial F^i}{\partial x^j} = \partial_j F^i$$

Definition 1.3 (Charts). Let M be a set. An m dimensional *chart* (U, φ) on M is a open subset $U \subset M$ together with a map $\varphi : U \to \mathbb{R}^m$ such that $\varphi(U)$ is an open subset of $\mathbb{R}^m$ and φ is bijective from U to $\varphi(U)$. The set U is called the *chart domain* and φ is called the *coordinate map*.

Definition 1.4 (Compatible Charts). Two charts (U, φ) and (V, ψ) on M are said to be *compatible* iff the subsets $\varphi(U \cap V)$ and $\psi(U \cap V)$ are open and the transition map

$$\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$$

is a diffeomorphism. Note that charts such that $U \cap V = \emptyset$ are always compatible.

Remark 1.5. Note the following properties for sets X, Y with a map $f : X \to Y$ and $A, B \subset X$ and $C, D \subset Y$.

1. For any map, $f(A \cup B) = f(A) \cup f(B)$.
2. For an injective map, $f(A \cap B) = f(A) \cap f(B)$.
3. For an injective map, $f(A \setminus B) = f(A) \setminus f(B)$.
4. For a bijective map, $f(A^c) = [f(A)]^c$.

If we denote the preimage of a set $C \subset Y$ by $f^{-1}(C) = \{x \mid f(x) \in C\}$, then:

1. $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$,
2. $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$,
3. $f^{-1}(C \setminus D) = f^{-1}(C) \setminus f^{-1}(D)$,
4. $f^{-1}(C^c) = [f^{-1}(C)]^c$.

Given a chart (U, φ) and a point $p \in U$, then the *coordinates* of p in the given chart are written as $\varphi(p) = (u^1, u^2, \dots u^m)$. The transition map $\varphi \circ \psi^{-1}$ between compatible charts is called a *change of coordinates*.

Definition 1.6 (Atlas). Let M be a set, then an m dimensional *atlas* on M is a collection of charts $\mathscr{A} = \{(U_\alpha, \varphi_\alpha)\}$ such that

1. U_α covers all of M, i.e. $\bigcup_{\alpha \in I} U_\alpha = M$, where I is an index set.
2. For each index $\alpha, \beta \in I$, the charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are compatible.

A chart (U, φ) is said to be compatible with an atlas $\mathscr{A}$ if it is compatible with all charts in $\mathscr{A}$.

Example 1.7. Consider M to be the set of all affine lines (as in lines not necessarily passing through origin) in $\mathbb{R}^2$.

$$M = \{\ell \mid \ell \text{ is an affine line in } \mathbb{R}^2\}$$

Let $U \subset M$ be the set of all non-vertical lines in M and $V \subset M$ be the set of all non-horizontal lines in M. Each line in U is described by the equation

$$y = mx + c.$$

Hence we let $\varphi : U \to \mathbb{R}^2$ be a map such that $\varphi(\ell) = (m, c)$ where $m, c \in \mathbb{R}$. Similarly, each line in V is described by

$$x = ny + b.$$

So we choose the map $\psi : V \to \mathbb{R}^2$ such that $\psi(\ell) = (n, b)$ where $n, b \in \mathbb{R}$. We see that (U, φ) and (V, ψ) are 2 dimensional charts on M. Notice that $A \coloneqq \varphi(U \cap V) = \psi(U \cap V) = \{(a, b) \in \mathbb{R}^2 \mid a \neq 0\}$ is open in $\mathbb{R}^2$, and the map $\psi \circ \varphi^{-1} : A \to A$ is defined by $\psi \circ \varphi^{-1}(m, c) = (1/m, -c/m)$ is smooth, and its inverse $\varphi \circ \psi^{-1}(n, b) = (1/n, -b/n)$ is also smooth. Hence, $\{(U, \varphi), (V, \psi)\}$ is an atlas on M.
Notice that if we reparametrize $m = \tan\theta$ and $n = \tan\phi$, then for $m = \tan(\theta + \pi)$ as well, hence after we advanced some steps in this 2 dimensional manifold, we return to the same point. This is characteristic of a Möbius strip!

Lemma 1.8. *Let $\mathscr{A} = \{(U_\alpha, \varphi_\alpha)\}$ be an atlas on a set M. If two charts (U, φ) and (V, ψ) are compatible with $\mathscr{A}$, then they are compatible with each other.*

Proof. For each chart $(U_\alpha, \varphi_\alpha)$, the sets $\varphi_\alpha(U_\alpha \cap U)$ and $\varphi_\alpha(U_\alpha \cap V)$ will be open, hence so will their intersection. The intersection is given by the following, by applying Remark 1.5 here.

$$\varphi_\alpha(U_\alpha \cap U) \cap \varphi_\alpha(U_\alpha \cap V) = \varphi_\alpha(U_\alpha \cap U \cap V)$$

Since $\varphi \circ \varphi_\alpha^{-1}$ is a diffeomorphism, it follows that

$$\varphi(U_\alpha \cap U \cap V) = (\varphi \circ \varphi_\alpha^{-1})(\varphi_\alpha(U_\alpha \cap U \cap V))$$

is an open set. Now, taking the union over all α we see that we obtain the open set

$$\varphi(U \cap V) = \bigcup_\alpha \varphi(U_\alpha \cap U \cap V)$$

A similar argument applies of $\psi(U \cap V)$. Now, the transition map $\psi \circ \varphi^{-1} : \varphi(U \cap V) \to \psi(U \cap V)$ will be smooth since for each α, the restriction to $\varphi(U_\alpha \cap U \cap V)$ is a composition of smooth maps $\varphi_\alpha \circ \varphi^{-1} : \varphi(U_\alpha \cap U \cap V) \to \varphi_\alpha(U_\alpha \cap U \cap V)$ and $\psi \circ \varphi_\alpha^{-1} : \varphi(U_\alpha \cap U \cap V) \to \psi(U_\alpha \cap U \cap V)$. Likewise, $\varphi \circ \psi^{-1} : \psi(U \cap V) \to \varphi(U \cap V)$ is also smooth.
Hence, charts (U, φ) and (V, ψ) are compatible with $\mathscr{A}$, are compatible with each other as well. □

Theorem 1.9. *Given an atlas $\mathscr{A} = (U_\alpha, \varphi_\alpha)$ on M, let $\tilde{\mathscr{A}}$ be the collection of all charts (U, φ) compatible with $\mathscr{A}$. Then, $\tilde{\mathscr{A}}$ is also an atlas, and is the largest possible atlas containing $\mathscr{A}$.*

Proof. Its easy to see $\tilde{\mathscr{A}}$ contains $\mathscr{A}$, since each chart $(U_\alpha, \varphi_\alpha) \in \mathscr{A}$ will also be contained in $\tilde{\mathscr{A}}$ because the charts are compatible with the atlas containing them. By the above lemma, we conclude each chart in $\tilde{\mathscr{A}}$ is compatible with each other since they are compatible with $\mathscr{A}$, and hence $\tilde{\mathscr{A}}$ is an atlas. Now, let (U, φ) be a chart such that it is compatible with $\tilde{\mathscr{A}}$, then it is also compatible with $\mathscr{A}$ and hence $(U, \varphi) \in \tilde{\mathscr{A}}$. This shows that $\tilde{\mathscr{A}}$ cannot be extended furthur. □

Definition 1.10 (Maximal Atlas)**.** An atlas is said to be *maximal* if it is not contained within a larger atlas. Given an arbitrary atlas $\mathscr{A}$, one calls $\tilde{\mathscr{A}}$ (as in the above Theorem) the maximal atlas determined by $\mathscr{A}$.

Definition 1.11 (Manifold)**.** An m dimensional *manifold* is a set M, together with a maximal atlas $\mathscr{A} = \{(U_\alpha, \varphi_\alpha)\}$ with the following properties:

1. **(Second Countability)** M is covered by countably many charts in $\mathscr{A} = \{(U_\alpha, \varphi_\alpha)\}$, i.e. the index set I containing index α is countable.
2. **(Hausdorff Condition)** For any two distinct points $p, q \in M$, $\exists$ charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) in $\mathscr{A}$ such that $p \in U_\alpha, q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$.

1.2 Projective Spaces

Definition 1.12 (Real Projective Space)**.** The n dimensional *real projective space* $\mathbb{RP}^n$ is the set of all lines (1D subspace) in $\mathbb{R}^{n+1}$ through the origin. It may be regarded as the following quotient space:

$$\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{\mathbf{0}\}) / \sim$$

for the equivalence relation

$$\mathbf{x} \sim \mathbf{y} \iff \exists\, \lambda \in \mathbb{R} \setminus \{0\} \text{ such that } \mathbf{y} = \lambda \mathbf{x}$$

The equivalence classes of $\mathbf{x} = (x^0, x^1, \dots x^n)$ is under this relation is commonly denoted as

$$[\mathbf{x}] = (x^0 : x^1 : \dots x^n)$$

which are called *homogenous coordinates*. $\mathbb{RP}^n$ has the following standard maximal atlas $\mathscr{A}$

$$\mathscr{A} = \{(U_0, \varphi_0), \dots (U_n, \varphi_n)\}$$

defined as follows: For $0 \leq j \leq n$, let

$$U_j = \left\{(x^0 : \dots x^n) \in \mathbb{RP}^n \mid x^j \neq 0\right\}$$

be the set for which j-th coordinate is non zero, and let

$$\varphi_j : U_j \to \mathbb{R}^n \text{ such that } (x^0 : \dots x^n) \mapsto \left(\frac{x^0}{x^j}, \dots \frac{x^{j-1}}{x^j}, \frac{x^{j+1}}{x^j}, \dots \frac{x^n}{x^j}\right)$$

It is immediate that φ_j is a bijection from U_j to $\mathbb{R}^n$, with the inverse map

$$\varphi^{-1}(u^1, \dots u^n) = (u^1 : \dots u^j : 1 : u^{j+1} : \dots u^n)$$

Now, we want to verify this is an atlas. Clearly, U_j covers $\mathbb{RP}^n$ and $\varphi_j(U_j \cap U_i)$ and $\varphi_i(U_j \cap U_i)$ are open sets. Now, assuming $i < j$ without loss of generality, one computes the transition map to be:

$$\varphi_i \circ \varphi_j^{-1}(u^1, \dots u^n) = \left(\frac{u^1}{u^i}, \dots \frac{u^{i-1}}{u^i}, \frac{u^{i+1}}{u^i}, \dots \frac{u^j}{u^i}, \frac{1}{u^i}, \frac{u^{j+1}}{u^i}, \dots \frac{u^n}{u^i}\right)$$

This map, and its inverse are both smooth, hence a diffeomorphism. A later proposition in the next chapter will show that this atlas has the Hausdorff property. Hence, we have a manifold $\mathbb{RP}^n$ of n dimensions.

Remark 1.13. Geometrically, $U_j \subset \mathbb{RP}^n$ consists of the lines intersecting the affine hyperplane

$$H_j := \{\mathbf{x} \in \mathbb{R}^{n+1} \mid x^j = 1\}$$

and the map φ_j takes a line $\ell \in \mathbb{RP}^n$ to its unique intersection point with H_j, followed by the identification $H_j \cong \mathbb{R}^n$. Hence, the complement of U_j consists of all lines parallel to H_j. We may identify U_j^c with $\mathbb{RP}^{n-1}$. Thus, as sets, $\mathbb{RP}^n$ is a disjoint union.

$$\mathbb{RP}^n = \mathbb{R}^n \sqcup \mathbb{RP}^{n-1}$$

Inductively, we obtain a decomposition:

$$\mathbb{RP}^n = \mathbb{R}^n \sqcup \mathbb{R}^{n-1} \sqcup \dots \mathbb{R} \sqcup \mathbb{R}^0$$

where $\mathbb{R}^0 = \{0\}$. We can later talk about this as a decomposition into not just sets, but submanifolds.

1.3 Open Subsets

Let M be a set equipped with an m dimensional maximal atlas $\mathscr{A} = \{(U_\alpha, \varphi_\alpha)\}$.

Definition 1.14 (Open Subset)**.** A subset $U \subset M$ is an *open subset* of M iff for all charts in $\mathscr{A}$, the set $\varphi_\alpha(U \cap U_\alpha)$ is open.

Proposition 1.15. *Given $U \subset M$, let $\mathscr{B} \subset \mathscr{A}$ be any collection of charts whose union contains U. Then U is open iff for all charts (U_β, φ_β) in $\mathscr{B}$ the sets $\varphi(U \cap U_\beta)$ are open.*

Proof. The forward part of the proof is obvious, so we work on the backward part. We reserve β to index charts in $\mathscr{B}$. Suppose $\varphi_\beta(U \cap U_\beta)$ is open for all β. Let $(U_\alpha, \varphi_\alpha)$ be chart in maximal atlas $\mathscr{A}$. We have that

$$\begin{aligned}
\varphi_\alpha(U \cap U_\alpha) &= \bigcup_\beta \varphi_\alpha(U \cap U_\beta \cap U_\alpha) \\
&= \bigcup_\beta (\varphi_\alpha \circ \varphi_\beta^{-1}) \varphi_\beta(U \cap U_\alpha \cap U_\beta) \\
&= \bigcup_\beta (\varphi_\alpha \circ \varphi_\beta^{-1})(\varphi_\beta(U_\alpha \cap U_\beta) \cap \varphi_\beta(U \cap U_\beta))
\end{aligned}$$

Since $\mathscr{B} \subset \mathscr{A}$, all $\varphi_\alpha(U_\alpha \cap U_\beta)$ are open. Hence, the intersection with $\varphi_\beta(U \cap U_\beta)$ is open, and so is the preimage under the diffeomorphism $\varphi_\alpha \circ \varphi_\beta^{-1}$. Finally, we use that a union of open sets in $\mathbb{R}^m$ is open. □